

WEEK 1 TASK

DESIGNING A SCALABLE IoT ARCHITECTURE

Comprehensive Technical Design Report

Objective: Design a scalable IoT architecture capable of supporting a growing network of devices and sensors.

Deliverables: Network layout, device interconnections, communication protocols, data-flow paths, simulation plan, performance analysis, risk assessment and mitigation.

Simulation: Cisco Packet Tracer / GNS3 methodology

Report Date: 30 August 2026

Table of Contents

1. Introduction
2. Objectives and Scope
3. IoT Architecture Fundamentals
4. Network Topologies
5. Communication Protocols
6. Proposed Scalable Architecture
7. Network Layout and Addressing
8. Device–Gateway–Cloud Interconnections
9. Data Flow Design
10. Edge Computing Strategy
11. Security Architecture
12. Scalability Strategy
13. Simulation Environment
14. Simulation Scenarios and Results
15. Performance Metrics
16. Risk Assessment and Mitigation
17. Design Assumptions
18. Implementation Roadmap
19. Future Enhancements
20. Conclusion and References

1. Introduction

The Internet of Things (IoT) connects physical sensors, actuators, embedded controllers, gateways, communication networks, software services and cloud platforms. A practical IoT deployment must handle physical-world constraints while maintaining the reliability, security and manageability expected from modern information systems.

This report proposes a layered architecture suitable for smart buildings, campuses, environmental monitoring, agriculture or industrial sensing. The architecture separates sensing, edge processing, IP networking, messaging, cloud analytics and management. This separation allows individual layers to be expanded without redesigning the entire system.

The design uses a hybrid topology, local sensor clusters, one or more IoT gateways, an IP backbone, MQTT messaging and cloud-based storage and analytics. Edge processing is used to filter, aggregate and buffer data before it is forwarded upstream.

Scalability is demonstrated conceptually through device-count and message-rate scenarios. The report also defines measurable metrics such as throughput, delay, packet loss, resource utilization and recovery time. Numerical scenario values are explicitly planning/illustrative values and should be replaced by measurements from an executed simulator run.

Security is integrated throughout the architecture. Unique device identities, network segmentation, protected transport, access control, software-update procedures and monitoring are treated as core design requirements rather than optional features.

2. Objectives and Scope

The primary objective is to design an IoT network that can begin as a small pilot and grow toward hundreds or thousands of devices while maintaining acceptable performance and security.

Specific objectives are to compare common IoT topologies; evaluate communication protocols; design a device–gateway–cloud architecture; define addressing and VLAN segmentation; use edge processing to reduce upstream traffic; plan simulation scenarios; establish performance targets; and identify risks with mitigation strategies.

The scope includes logical architecture, network structure, communication, data flow, security, scalability and simulation methodology. Hardware procurement, detailed cloud billing, regulatory certification and production safety validation are outside the scope.

The base design assumes approximately 50 devices and a growth path toward 500–1,000 devices. It is intentionally vendor-neutral so that a suitable cloud provider, gateway hardware or sensor family can be selected later.

3. IoT Architecture Fundamentals

A scalable IoT system can be divided into device, edge, network, messaging, application and management/security layers. The device layer measures or controls physical conditions. The edge layer aggregates and preprocesses information. The network layer provides connectivity and segmentation. The messaging layer transports telemetry. Applications store, analyze and visualize data.

Layering reduces coupling. A sensor can use a low-power local connection while the gateway exposes MQTT upstream. An application can be replaced without replacing all sensors as long as the gateway and data contracts remain stable.

The architecture also supports graceful degradation. During a temporary cloud outage, the gateway can continue collecting data and buffer records. Local rules can continue to operate where appropriate, reducing dependence on continuous WAN connectivity.

This layered approach also simplifies troubleshooting because faults can be isolated to a device, local link, gateway, network, broker, application or storage layer.

4. Network Topologies

Star topology connects multiple devices to a central gateway. It is simple and easy to manage but creates a central dependency. Mesh topology provides multiple paths and can improve resilience, although it introduces additional routing complexity and may consume more device energy.

Tree topology provides hierarchical expansion and is useful for buildings or campuses with zones. However, a failure near the top of the hierarchy can affect an entire branch. Hybrid topology combines several approaches and is therefore recommended for the proposed design.

The recommended network uses local star or mesh sensor clusters around gateways and Ethernet/IP connectivity from gateways toward the core. This allows a new sensor group or building zone to be added without redesigning the entire backbone.

Topology selection should also consider power, distance, interference, maintenance skill, installation cost and required availability. The best topology is application-dependent rather than universally fixed.

5. Communication Protocols

MQTT is selected as the primary telemetry protocol because it uses lightweight publish/subscribe messaging. Sensors publish to topics while authorized applications subscribe. This decouples producers and consumers and supports multiple data consumers without direct sensor-to-application connections.

HTTPS is appropriate for web dashboards, administrative APIs and configuration services. CoAP is an optional choice for constrained devices requiring a compact request/response protocol. Wi-Fi is suitable for powered devices, while low-power IEEE 802.15.4-class technologies can serve battery-operated sensor networks.

Ethernet is preferred for gateway uplinks because gateway traffic is more sustained than individual sensor traffic. IP networking provides routing and segmentation, while an IPv6-ready design preserves a long-term path for large addressing requirements.

Protocol selection should be based on payload size, energy consumption, latency, reliability, interoperability and security support rather than popularity alone.

6. Proposed Scalable Architecture

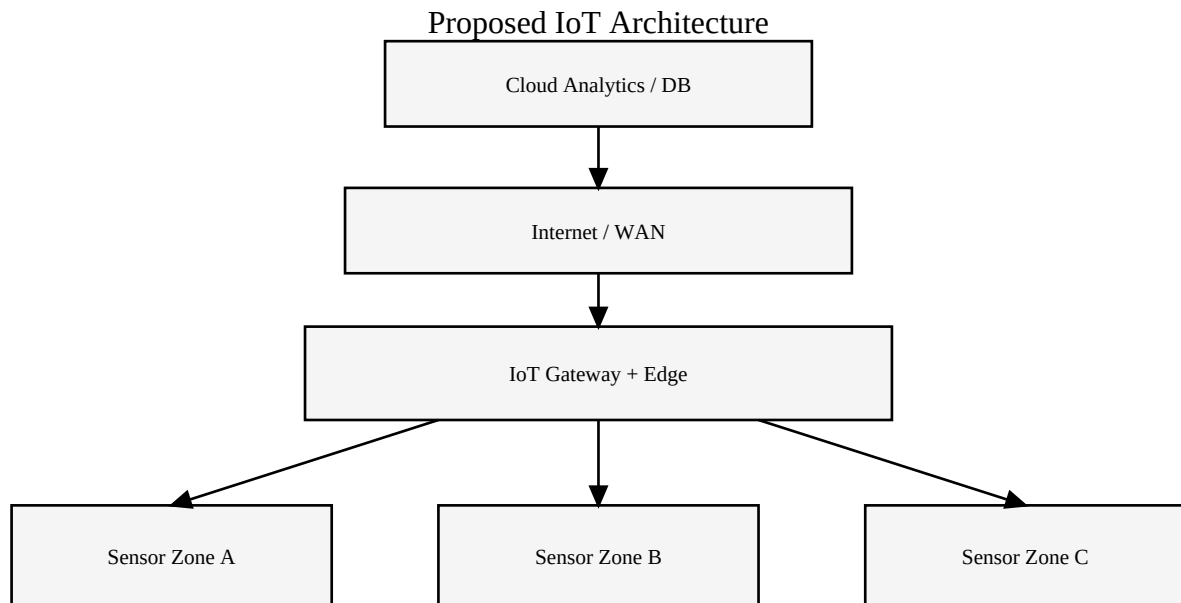


Figure 1. Layered scalable architecture

The architecture has sensing zones, an edge/network zone and a cloud/application zone. Sensors send measurements to a local gateway. The gateway performs validation, filtering, aggregation, buffering and protocol translation before publishing normalized telemetry upstream.

The messaging layer decouples devices from analytics applications. As the fleet grows, additional gateways can be introduced by zone. At larger scale, broker and application services can be clustered or horizontally scaled.

The architecture intentionally avoids a single point of dependence for every function. A local gateway can continue basic operation during a cloud outage, while redundant gateways can be introduced where higher availability is required.

7. Network Layout and Addressing

The laboratory network uses private IPv4 addressing with an IPv6-ready logical design. VLANs separate IoT devices, gateways, servers and administration. Segmentation reduces broadcast traffic and limits lateral movement.

VLAN	Purpose	Subnet	Devices
10	IoT Sensors	192.168.10.0/24	Sensors/controllers
20	IoT Gateways	192.168.20.0/24	Gateway nodes
30	Servers	192.168.30.0/24	MQTT/database/monitoring
40	Management	192.168.40.0/24	Admin workstation
50	Guest/Test	192.168.50.0/24	Temporary devices

Example infrastructure addresses are 192.168.10.1, 192.168.20.1, 192.168.30.1 and 192.168.40.1 for the respective routed VLANs. Infrastructure addresses should be reserved. Sensor addresses can use DHCP reservations or an onboarding service.

When a segment becomes crowded, add a new zone or VLAN rather than indefinitely expanding one broadcast domain. This improves fault isolation and policy control.

8. Device–Gateway–Cloud Interconnections

A telemetry transaction starts with a sensor measurement. The controller validates the reading and adds a timestamp and device identifier. The gateway receives multiple streams and applies filtering, aggregation and buffering before publishing an MQTT message.

Component	Input	Processing	Output
Sensor	Physical signal	Sampling/calibration	Measurement
Controller	Measurement	Validation/timestamp	Record
Gateway	Multiple streams	Filter/aggregate/buffer	MQTT telemetry
Broker	MQTT messages	Topic routing/QoS	Subscriber streams
Analytics	Telemetry	Rules/statistics	Alerts/insights
Database	Records	Storage/indexing	Historical data

Buffering is important because WAN connectivity can fail temporarily. The gateway can queue data and forward it when connectivity returns. The data contract should include device ID, timestamp, measurement type, value, unit and quality/status information.

9. Data Flow Design

Telemetry and Control Flow

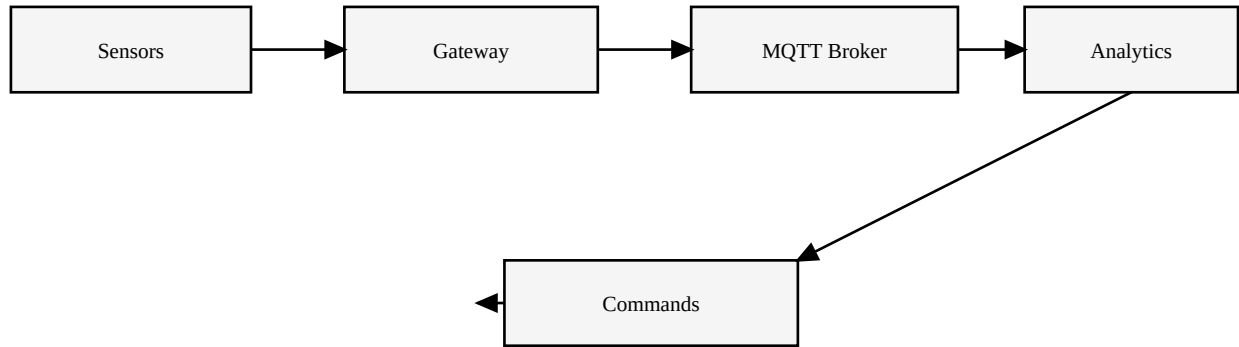


Figure 2. Telemetry and controlled command paths

Telemetry follows sensor → gateway → MQTT broker → analytics/database → dashboard. Control commands follow the reverse path only after authentication and authorization.

Suggested topics are `site/{site_id}/zone/{zone_id}/device/{device_id}/telemetry` and a parallel command branch. Predictable topic structures make permissions, monitoring and troubleshooting easier.

Validation should occur at the gateway and application boundary. Malformed records, unknown devices and impossible values should be rejected or quarantined.

10. Edge Computing Strategy

Edge computing moves selected processing close to the devices. The gateway performs filtering, aggregation, buffering and local decision-making.

Function	Example	Benefit
Filtering	Reject impossible readings	Less traffic
Aggregation	Average multiple samples	Lower message rate
Buffering	Store during WAN outage	Resilience
Local rules	Threshold detection	Lower latency
Translation	Local protocol to MQTT	Interoperability

If 500 devices report once every 10 seconds, the raw event rate is approximately 50 messages per second. If edge processing reduces transmitted events by 60%, the upstream event rate becomes approximately 20 messages per second. This is a planning calculation, not a measured result.

Edge rules should be versioned, monitored and updateable. Safety-critical actions require additional engineering validation before deployment.

11. Security Architecture

Security is implemented in layers. Each device should have a unique identity. Network segmentation restricts unnecessary communication. MQTT should use authentication and protected transport in production. Cloud services should apply least privilege, logging and backups.

Threat	Effect	Mitigation
Unauthorized device	False data/access	Identity and registration
Credential theft	Unauthorized messaging	Rotation, certificates/TLS
Eavesdropping	Data disclosure	Protected transport
Gateway compromise	Large disruption	Hardening, updates, monitoring
Unsafe firmware	Device compromise	Authorized updates
Message flood	Broker congestion	Rate limits and monitoring
Data loss	Historical loss	Backups and replication

Security must cover onboarding, operation, maintenance, updates and retirement. A device that cannot be uniquely identified or securely updated presents long-term risk.

12. Scalability Strategy

Growth	Bottleneck	Upgrade
50 → 200 devices	Gateway capacity	Add gateway/split zones
200 → 500	Broker connections	Tune or cluster broker
500 → 1,000+	Network/management	Multiple zones + automation
Higher sample rate	Bandwidth/storage	Aggregation/compression
More analytics	Compute	Horizontal cloud scaling
More sites	WAN dependency	Regional gateways

Scalability should be horizontal wherever practical. Instead of relying on one very large gateway, distribute devices across zones. Instead of one messaging server handling all workloads, introduce clustering or partitioning when required.

Monitoring should track CPU, memory, network utilization, broker connections, throughput, queue depth and database write latency. A planning goal is to keep sustained utilization below approximately 70% of tested capacity, leaving headroom for bursts.

13. Simulation Environment

Cisco Packet Tracer or GNS3 can model the proposed logical network. The simulation should contain IoT devices, gateway/router nodes, a switch, a server representing the messaging/cloud layer and an administration workstation.

Component	Base Quantity	Purpose
IoT devices	50	Generate sensor traffic
Gateway	1	Aggregate traffic
Switch	1	LAN connectivity
Router	1	Routing/segmentation
Server	1–2	MQTT/cloud representation
Admin PC	1	Monitoring/configuration

Procedure: build the topology; configure IP addressing; verify connectivity; configure VLANs; introduce traffic; record results; increase device count; increase message frequency; then add a second gateway and compare results.

Simulator-specific application-layer behavior can differ from production hardware. Therefore, the simulation should be treated as validation of topology, traffic growth, segmentation and bottleneck behavior.

14. Simulation Scenarios and Results

Note: The following are illustrative planning values. They are not claimed measurements from a completed Packet Tracer/GNS3 execution. Replace them with actual readings before final submission.

Scenario	Devices	Approx. Msg/s*	Expected Load	Expected Status
S1 Pilot	50	5	Low	Stable
S2 Growth	100	10	Low–Moderate	Stable
S3 Medium	250	25	Moderate	Stable with monitoring
S4 Large	500	50	Moderate–High	Scale if needed
S5 Stress	1,000	100	High	Multiple gateways/broker

*Assumes one event per device every 10 seconds. Payload and protocol overhead are excluded from this simple event-rate calculation.

For each scenario record average and 95th-percentile delay, packet loss, throughput, gateway utilization where available, broker connections, queue depth and recovery time after a simulated failure.

Expected qualitative behavior is that the pilot and medium scenarios remain stable while high device counts increase gateway and broker contention. Distributed gateways and scalable messaging should reduce contention.

15. Performance Metrics

Metric	Definition	Interpretation
Throughput	Processed messages/sec	Must exceed peak input
Latency	Event-to-application delay	Track average and p95
Packet loss	Failed/total packets	Near zero on controlled LAN
Gateway utilization	CPU/memory/network	Avoid sustained saturation
Availability	Operational time	Higher is better
Recovery time	Restore time after failure	Set application objective
Data freshness	Age of latest valid reading	Match use case

Averages can hide short spikes. Tail latency, queue depth and maximum utilization should also be recorded. Tests should include steady traffic and burst traffic because many devices may reconnect at the same time after a network or power interruption.

Performance targets depend on the application. Monitoring may tolerate seconds of delay, while local control may require much faster response and should therefore execute at the edge.

16. Risk Assessment and Mitigation

Risk	Likelihood	Impact	Level	Mitigation
Device compromise	High	High	Critical	Identity, hardening, updates, monitoring
Gateway failure	Medium	High	High	Redundancy, backups, health checks
Network congestion	High	Medium	High	QoS, aggregation, segmentation
Cloud outage	Medium	High	High	Edge buffering/local operation
Credential leakage	Medium	High	High	Rotation, least privilege, TLS
Data loss	Low–Medium	High	Medium/High	Replication and backups
Power failure	Medium	Medium	Medium	UPS where appropriate
Configuration error	Medium	Medium	Medium	Version control/change management

The highest-priority risks are device compromise, gateway failure and network congestion because they can affect many devices simultaneously. The proposed controls address identity, resilience, segmentation, monitoring and capacity.

17. Design Assumptions

1. Initial deployment contains approximately 50 sensors and is intended to scale toward 500–1,000 devices.
2. Devices have unique identifiers and suitable local connectivity.
3. Gateways provide temporary storage for buffering.
4. Switches and routers support VLANs and IP routing.
5. MQTT is available as the primary telemetry mechanism.
6. Cloud connectivity may temporarily fail.
7. Administrators are authorized to manage the infrastructure.
8. Simulation values depend on the chosen simulator and host machine.
9. Production capacity must be confirmed through hardware load testing.
10. The prototype is not assumed to be a safety-critical control system.

These assumptions keep the design suitable for academic simulation while preserving a path toward practical deployment.

18. Implementation Roadmap

Phase	Activities	Output
1 Requirements	Define devices, sampling and security	Requirements specification
2 Prototype	Build 10–50 device model	Working prototype
3 Network	VLANs, routing and gateway access	Segmented network
4 Messaging	MQTT topics, access and QoS	Telemetry pipeline
5 Edge	Filtering, aggregation, buffering	Reduced upstream load
6 Simulation	Increasing device/message scenarios	Performance dataset
7 Security	Access, update and logging tests	Security checklist
8 Scale-out	Add gateways/broker capacity	Validated scalable design

A phased approach reduces risk. Connectivity and data quality should be validated before load is increased. Security controls should be applied before exposure to untrusted networks. Scaling decisions should follow measured bottlenecks.

19. Future Enhancements

Enhancement	Purpose	Benefit
IPv6 / dual stack	Large address space	Long-term flexibility
MQTT broker cluster	Distribute messaging	Availability/throughput
Cloud orchestration	Elastic services	Automated scaling
Digital twins	Model device state	Improved monitoring
ML anomaly detection	Find unusual behavior	Earlier detection
OTA updates	Remote maintenance	Faster patching
Zero-trust controls	Continuous authorization	Reduced lateral movement
Time-series optimization	Efficient storage	Faster queries

Future improvements should be selected according to measured requirements. If gateway CPU is the bottleneck, add gateways. If broker capacity is limiting, cluster or partition messaging. If storage is dominant, aggregate data and optimize retention.

20. Conclusion and References

This report presents a scalable IoT architecture based on modular layers, hybrid connectivity, edge processing, MQTT messaging, segmented IP networking and cloud analytics. The design can grow from a 50-device pilot toward hundreds or thousands of devices through horizontal expansion of gateways, messaging infrastructure and application services.

The key principle is to avoid concentrating every function in one component. Edge gateways reduce upstream traffic and provide resilience during WAN interruptions. VLANs and access controls isolate IoT devices. MQTT provides decoupled messaging. Cloud services provide scalable storage and analytics.

The simulation methodology provides a practical way to demonstrate scalability. Device count and message rate should be increased in controlled stages while delay, throughput, packet loss, utilization and recovery are recorded. The illustrative results in this report should be replaced with actual simulator measurements before submission.

References

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